

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, suspension	Molecules Electroporated	Dextrans, various proteins
Species Used	Human, red blood cells		

Before the Pulse

Cell Growth Medium	Harvested from whole blood	Growth Phase at Harvest	Not given
		Pre-pulse Incubation	Held on ice in Phosphate Buffered Saline
Wash Solution	Isotonic Phosphate Buffered Saline (PBS)		

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature	0 or 25 °C		
Electroporation Medium*	20 mM Phosphate Buffered Saline	Cuvette Gap	0.2 cm
Cell Density	10 (6) to 10 (7) cells / ml	Voltage	0 to 2.5 kV
Volume of Cells	0.4 ml	Field Strength	0 to 12.5 kV/cm
DNA Concentration	10 (-5) to 10 (-4) M dextran	Capacitor	25 µF
DNA Resuspension Buffer	Protein in Phosphate Buffered Saline	Resistor	(Pulse Controller) not used.
Volume of DNA	Not given		
		Time Constant	1 to 2 msec

After the Pulse

Outgrowth Medium Phosphate Buffered Saline

Outgrowth Temperature	0 °C	Relevant Publications and/or Comments Note: exponential values designated in parentheses. **It is NOT RECOMMENDED to use high voltage with out the Pulse Controller. PBS: 1x = 8g NaCl, 0.2g KCl, 0.2g KH ₂ PO ₄ , 1.15g Na ₂ HPO ₄
Length of Incubation	0 to 5 hr.	
Selection Method or Assay Used	Flow cytometry, detection of fluorescent molecules	
Electroporation Efficiency	up to 90% of cells exhibit uptake	
Per Cent Survival	25 to 100%	

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Survey Number

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